Vector DBs
Powering the AI Revolution

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Machine Learning & Analytics
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Vector DBs Overview

What are Vector DBs

- Indexes, stores, and provides access to structured or unstructured data alongside its **vector embeddings**
- Vector embeddings are a prevalent data format used in **machine learning models (image/word -> embeddings)**
- Allows users to find and retrieve similar objects (rooted in semantic relevance) quickly at scale

Structured Data (SQL)

Relational DBs

Table

Unstructured Data (NoSQL)

Traditional Non-Relational DBs, Data Lakes, Data Warehouses

Key-value	Document	Column-based
Graph	In-memory	

Vector Databases

Embeddings & Metadata

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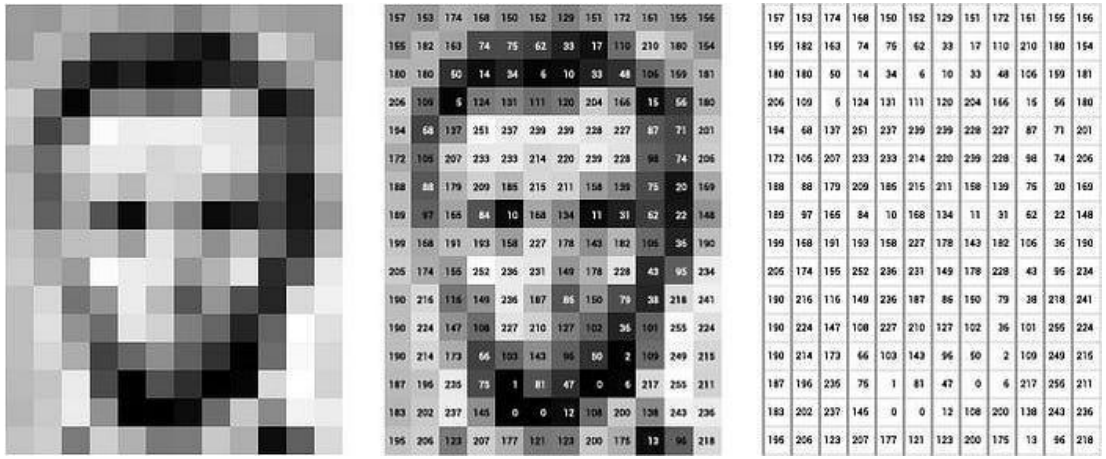

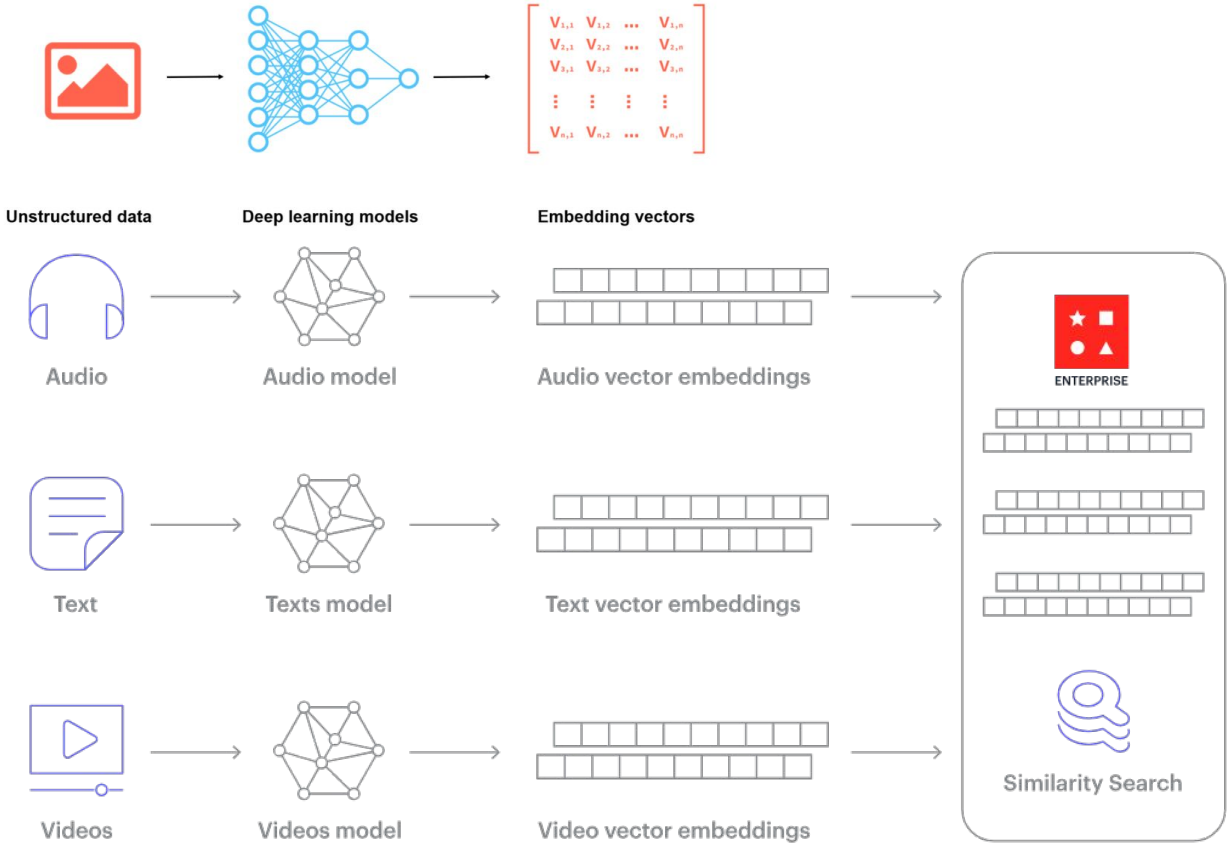

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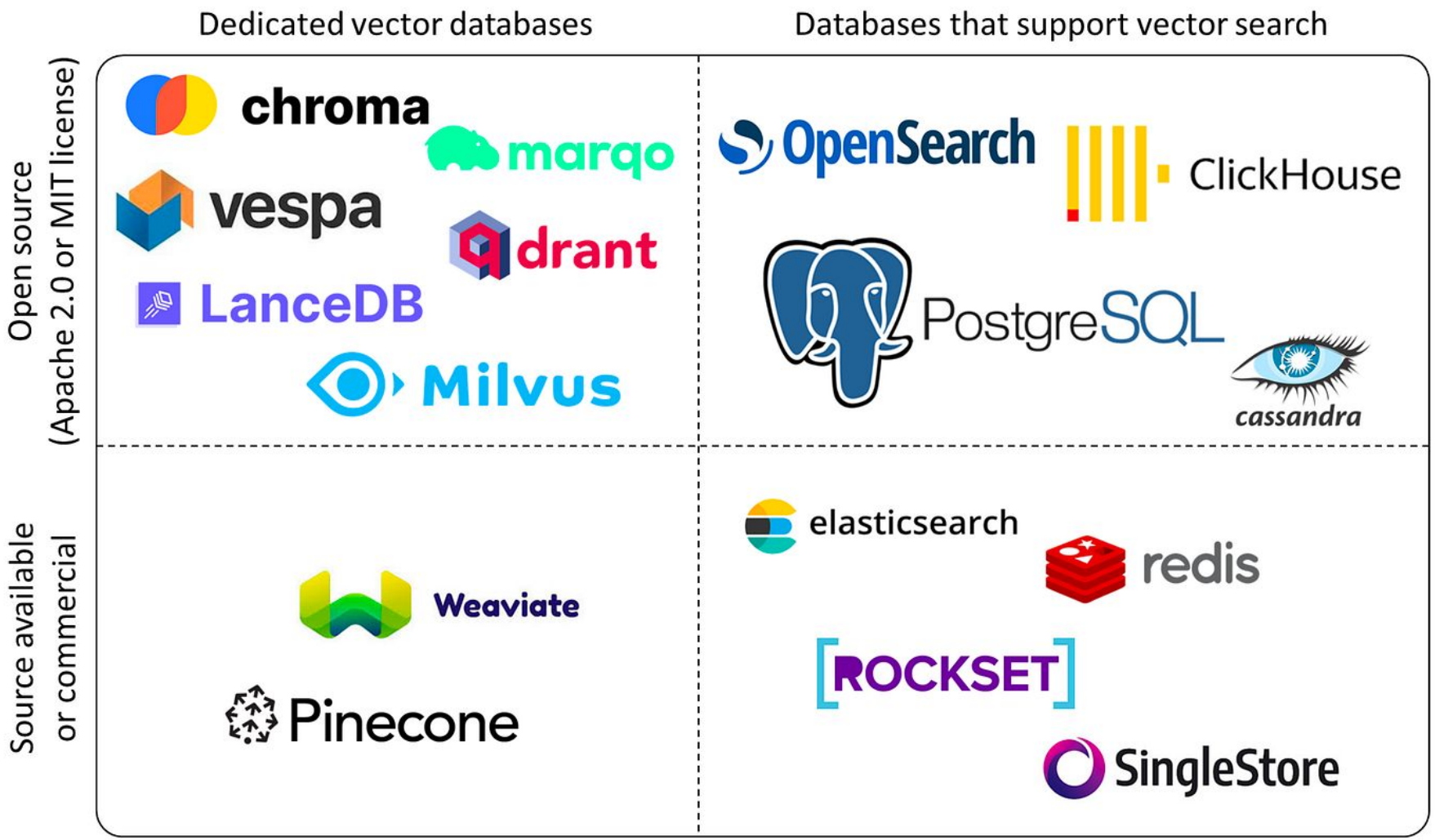
Providers



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Similarity Measures

While regular databases search for exact data matches, vector databases look for **the closest match** using specific measures of similarity

The basic rule of thumb in selecting the best similarity metric is to **match it to the one used to train your embedding model**

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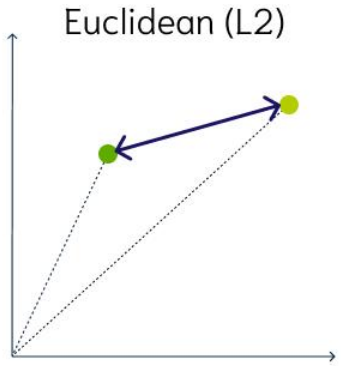
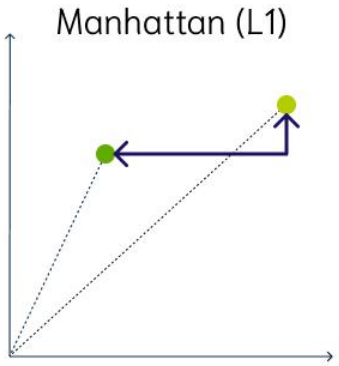
$$\sum_{i=1}^{n} (x_i - y_i)^2$$

"Shortest distance"

Takes into account:
- Magnitude of vectors
- Direction of vectors





$$\sum_{i=1}^{n} |x_i - y_i|$$

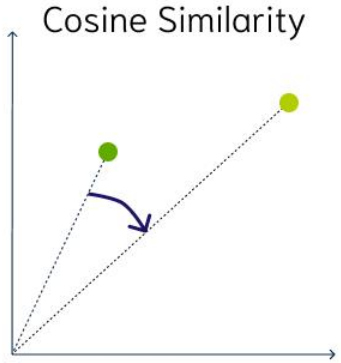
"Taxicab distance"

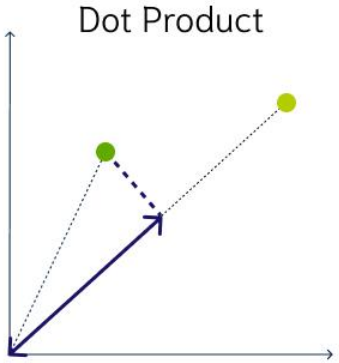
Takes into account:
- Magnitude of vectors
- Direction of vectors

$$\mathrm{Cos}(A,B) = \frac{A \cdot B}{||A||\,||B||}$$

"Similar vectors point in the same direction"
"Angle between 2 vectors"

Takes into account:
- Direction of vectors





$$A \cdot B = \sum_{i=1}^{n} A_i B_i$$

Takes into account:
- Magnitude of vectors
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Use Cases

Image and Video Recognition

Image/Video -> Image/Video/Label

E.g. User pins an image of a coastal sunset, the system can search through its vector database to suggest similar images, perhaps other beach landscapes or sunsets, enhancing content discovery and user engagement.

Natural Language Semantic Search

Retrieve semantic similar texts/documents

E.g. Customer support chatbot system, customer queries are transformed into vectors using embeddings. When a user asks, “How do I reset my password?” the vector database can identify semantically similar queries like “Steps for password change” to provide a relevant response even if the exact phrasing isn’t in the system.

Recommendation Systems

Recommend similar products based on user preferences

E.g. Netflix, when a user watches a psychological thriller starring a particular actor, the vector database can suggest other movies in the same genre or films with the same actor, offering a tailored viewing experience.

Anomaly/Fraud Detection

Identify anomaly by comparing input with vectors of stored anomalies

E.g. At an airport, each passenger’s face is captured and converted into a vector. When a passenger approaches the security check, their face is matched against a vector database of known criminals or persons of interest, ensuring rapid threat detection.

Autonomous Driving

Decide on action to take based on vectors of stored sensor data

E.g. Index and search sensor data from autonomous vehicles, allowing the system to quickly identify similar sensor readings and make real-time decisions based on the data. This can be useful for applications such as object detection and tracking, and path planning.

Drug Discovery and Genomics

Search similar compounds or genetic patterns

E.g. Chemical compounds are represented as high-dimensional vectors in a pharmaceutical research lab. When researchers identify a compound promising in treating a specific disease, the vector database can find other compounds with similar structures or properties, potentially leading to more efficient drug discovery processes

CLIP Model
Similarity search by leveraging VectorDB

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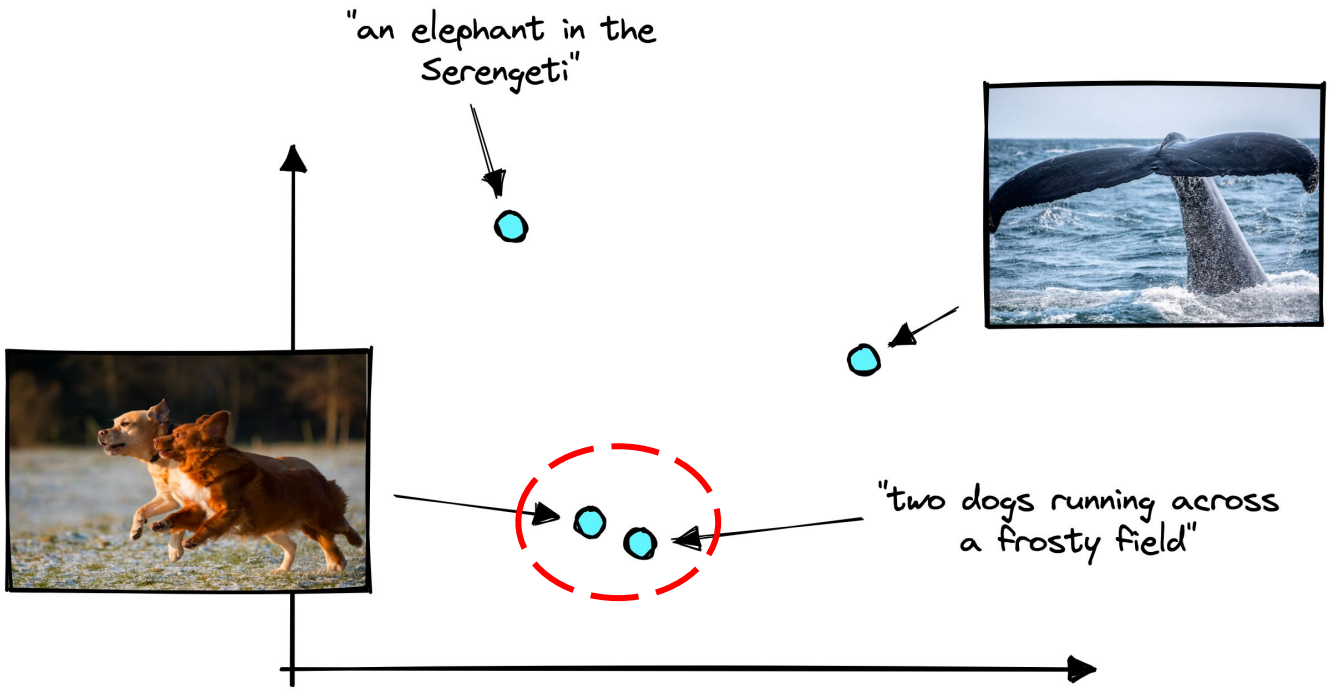
CLIP Model Overview

CLIP is the first **multimodal** (in this case, vision and text) model tackling computer vision and was recently released by OpenAI on January 5, 2021.

The multi-modal nature of CLIP is powered by two encoder models trained to "**speak the same language**".

This is done by **encoding similar concepts in text and images into similar vectors**



The *text* "two dogs running across a frosty field" would output a vector similar to an *image* of two dogs running across a frosty field

Used in OpenAI's DALL·E 2

CLIP is used to encode the textual semantics to their visual representations. It then generate images based on these representations using GLIDE Diffusion model.

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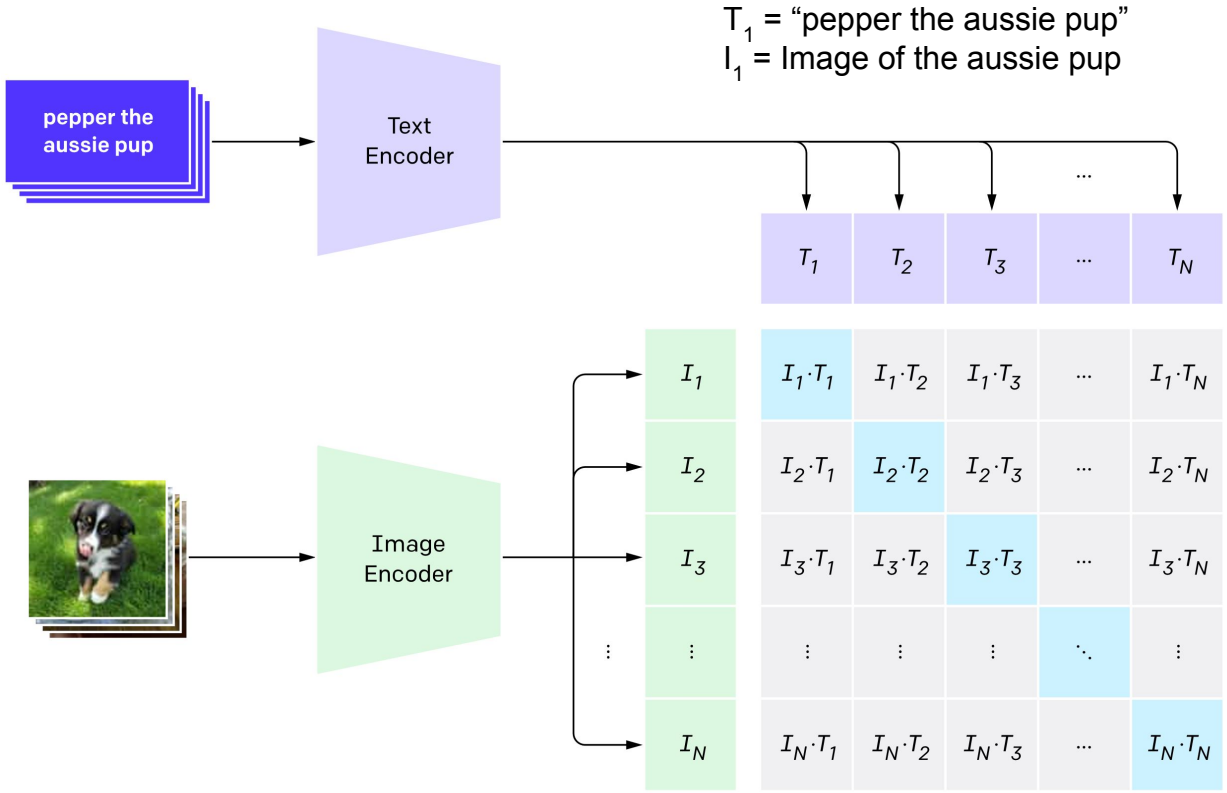
CLIP Model
Training Architecture

1. Contrastive pre-training



Text encoder is a transformer model;
Image encoder has 2 variants, Resnet or ViT
Output of each encoder is a vector with 512 dimensions

CLIP model was trained on 400 million (Text, Image) pairs for 32 epochs

Forward propagation

T = text ; I = image
Size of each batch, **N** = 32,768 (T, I) pairs

For each batch,
- T -> Text Encoder -> Text embedding
- I -> Image Encoder -> Image embedding

Objective function – Contrastive Loss

Each (T,I) pair should be close to each other in the vector space
- Maximise cosine similarity of I_jT_j

None (T,I) pairs should be far from each other in the vector space
- Minimise cosine similarity of I_jT_x where $\{x \in N \mid x \neq j\}$

Graphically, to maximise the cosine similarity of the "diagonal blue boxes" and minimize the "rest of the gray boxes

Backpropagation

Update the Text and Image Encoder weights by backpropagation

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CLIP Model
Training Architecture

> Note: The CLIP model trained on 400,000,000 labeled images. The training process took 30 days across 592 V100 GPUs. This would have cost **$1,000,000** to train on AWS on-demand instances!

CLIP Model
Notable Achievements

Using already-labelled data

- CLIP learns from text-image pairs that are already **publicly available on the internet**
- Existing state-of-the-art computer vision algorithms require significant human effort in labelling

Generalization

- Existing models tends to be good for the specific thing they were trained to do
  - Trained on benchmark dataset and evaluated on benchmark dataset
  - Gap in real-world vs benchmark performances
- CLIP not trained on benchmark data
  - Benchmark performance are **representative of real-world**

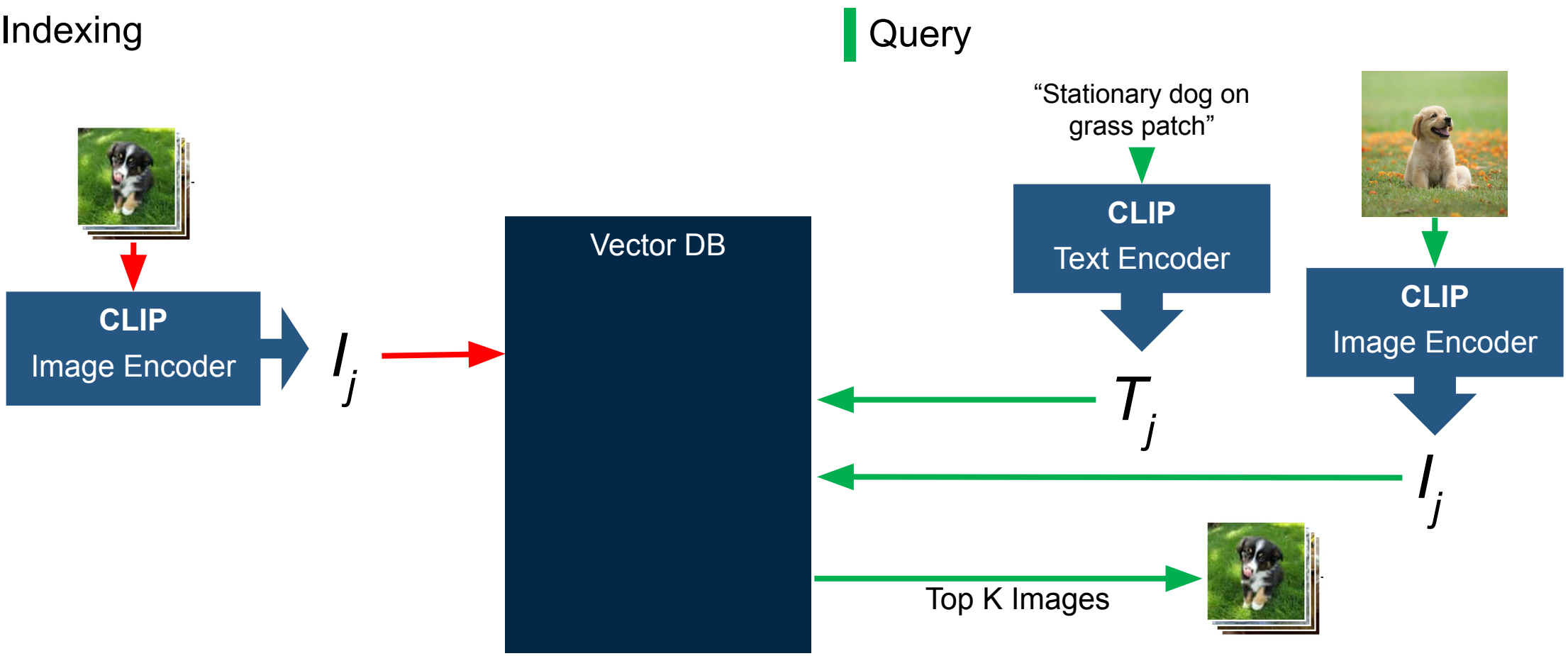
Zero-shot Image Classification

- Existing models have a finite number of labelled classes (ImageNet 1000 classes)
  - Must build new a dataset and finetune model to perform other tasks
- CLIP can be used **out of the box** for wide variety of image classification tasks
  - On-par/beats task-specific models
  - Best CLIP model outperformed the best ImageNet model on 20 out of the 26 datasets that were tested by the team

Multimodal (Text and Image)

- Ability to map vector representations of text and image on the **same vector space**
  - Similar representations close together
  - Dissimilar representations far apart

CLIP Model + Vector DB Similarity Search Architecture



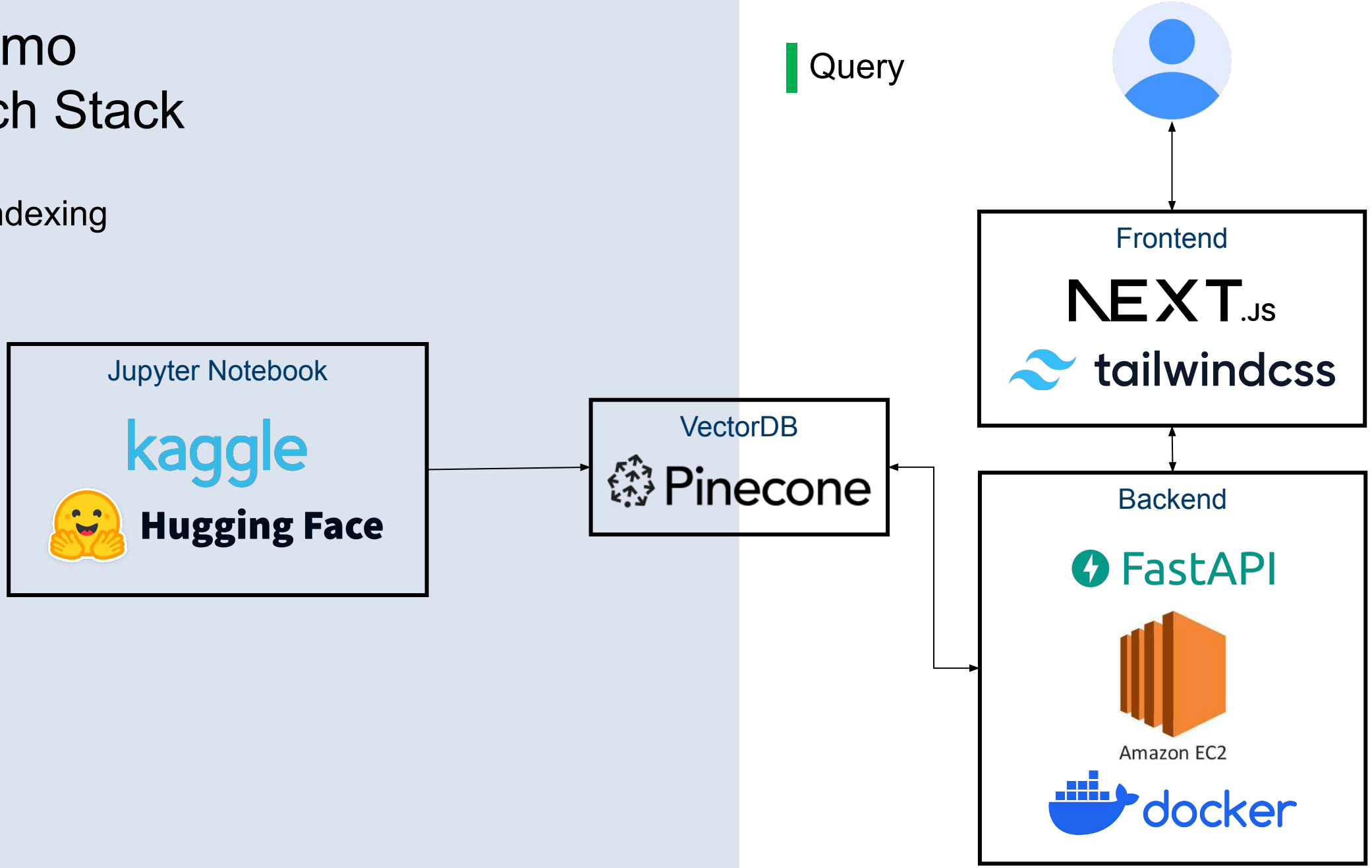
Since CLIP model uses cosine similarity as objective function, we will use it as our VectorDB metric.

Demo

Demo
Tech Stack

Indexing

Query



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Demo
Scenarios

	Scenarios
1	Semantic search (Monster, Cute)
2	Text – Representation (Dragon, Lizard, Neigh, Oink, Squishy, Boxer, Kungfu)
3	Text – Exact (Squirtle, Hitmonchan)
4	Images

Main Takeaways

Lessons

1. CLIP model allows us to **map image and text to the same vector space**
2. CLIP model is excellent at **zero-shot learning**
3. VectorDB allows us to store & index embeddings which allows us to quickly **search for closest vectors**

Further Exploration

More to learn…

- Fine-tuning CLIP to fit our use case
- Combination of VectorDB with LLMs (e.g. Llama 2) for Retrieval Augmented Generation (RAG)
  - Using Langchain to allow for in-context learning

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Thank you!

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